

What's On Offer?

Along with the obvious advantage of the PCIe bus, these boards offer a gamut of features previously unheard of: if you go out to buy a PCIe motherboard, expect to be bombarded with feature-rich boards. We take you through the best of what you can expect from these boards.

Video

The latest PCIe chipsets from Intel, the 915, 925 and their variants, come with an on-board Intel Graphics Media Accelerator, which has Direct-X 9 support, and is stipulated to match the performance of the GeForce FX5200. The 915G and 915GV chipsets also have native support for HDTV and widescreen LCDs, with support for multiple monitors. The AMD solutions from nVidia do little to better this, what with the nForce4 chipset shipping without any onboard video at all!

PCI: The Evolution Path	
Version	Technology
PCI	Intel developed the 32-bit PCI bus, which runs at 33 MHz with a transfer speed of 133 MB/s
PCI-X	The 64-bit version of the PCI bus, running at 33 MHz with a transfer speed of 133 MB/s
PCI-X	64-bit version running at 66MHz with speeds up to 512 MB/s
PCI-X	64-bit version running at 133MHz with speeds up to 1 GB/s
PCI-X 2.0	4.3 GB/s
The reason PCI-X is not implemented in desktops is the extremely high cost of manufacture of PCI-X boards	

ATi Radeon Xpress200/200P chipsets do much better, with their on-board video solution having 128 MB of memory, matching the performance of a Radeon X300. It also has built-in dual-display support (Analogue and DVI) along with SURROUNDVIEW, which is essentially support for three or more monitors.

Both Via K8T890 and K8T890PRO as well as SiS 756 and 965 chipsets do not feature any onboard video. So on this front, ATi leads the pack.

Audio

This is where the Intel chipsets emerge as clear winners. With on-board support for eight-channel Hi-Definition audio with a 192 kHz sampling rate, which can play everything from High Definition Dolby DTS to vanilla CDs without problems. It even supports up to 15 simultaneous audio streams.

The various AMD-based PCIe chipsets support eight-channel audio, but this is not High Definition. In fact, the Via K8T890 only has integrated AC-97 six-channel audio, though it does come with an eight-channel audio PCI controller.

The much-touted Soundstorm integration into nForce4 did not materialise, and we get the simple eight-channel integrated audio, which again, is not high-definition. Likewise with ATi and SiS. The Gigabyte K8NXP-9, though, improves on the nForce4 chipset and provides six programmable audio jacks. No High-Definition audio, though.

Storage And Memory

Every PCIe chipset comes with at least four SATA connections, with the Gigabyte K8NXP-9 offering eight. All offer ATA connections and most of them also have the capability to be configured in various RAID flavours: nForce4, supports 0, 1, 0+1

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RAID, and they also allow you to change the RAID configuration on the fly.

Another cool piece of technology that has made its way to today's motherboards is CQ, or Native Command Queuing. This technology enables a hard disk to decide the order of reading data to minimise data throughput time—so you can watch a movie even when some encoding is going on in the background, without stressing out your hard disk.

The nForce4 and Intel chipsets support this technology; however, neither the ATi Xpress200 nor the VIA or SiS chipsets support this feature.

It should be noted here that not all these boards support DDR2. For example, the VIA K8T890 series supports only DDR memory. The Intel 915GV and 915G support both DDR and DDR2. Note that the 915GV Express chipsets support only the latest LGA775 socket Pentium 4s.

Network Support And Other Connections

All these chipsets provide a variety of connectivity options. Each of these PCIe chipsets provides at least eight USB ports and one FireWire, with the nForce4 providing 10 USB and two FireWire ports. Apart from this, it also supports Gigabit LAN.

Interestingly, the Gigabyte K8NXP-9 offers two on-board GigabitLAN ports. One is an nForce 4, and the other, added by Gigabyte, utilises a PCIe x1 link. This apart, the motherboard ships with an additional PCI Wireless LAN card as well! The card is a Gigabyte addition and is not on-board. The ASUS A8V-E Deluxe does feature an on-board wireless LAN conforming to the 802.11g standard.

For gamers who will be upgrading anyway, ATi and nVidia have released their versions of the PCIe x16 graphics cards, which are available without having to pay a premium!

Unlike ATi, though, which launched native PCIe cards, nVIDIA's solutions use a PCIe-to-AGP bridge and have merely offered bridged versions of their current AGP 8X cards. However, the performance difference due to the bridge is negligible and you can enjoy playing your favourite games on high-end cards such as the 6800 GT and the Radeon X850.

In The Future

The sheer convenience of the serial connections offered by PCIe and the bandwidth available could change the motherboard manufacturing process entirely by giving greater flexibility in terms of component placements. We might see modular PCs where the I/O system is removable from the processor and could connect to it via PCIe sockets.

This does not signal the death of USB and FireWire, though, because by the time the above scenario is realised, they would be so integrated with our lives that even the modular PCs would need to provide legacy support for them; just as PCIe boards provide support for the older generation PCI cards.

Despite some glitches like non-compatibility with AGP, PCIe technology is remarkable. It has tremendous industry support, and not without reason. The bus is well on its way. ☑

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